

Predicting Systemic Delivery Risk in Supply Chains:

A Graph Attention Network Approach

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1. Introduction

In today's globally interconnected economy, the supply chain has evolved from a single linear pipeline to a series of multiple, complex, and interdependent networks. A single disruption, such as a late delivery, could trigger a ripple effect in the system and result in volatile consequences. The key to effectively mitigate recurring disruptions is not only to detect delays, but also to understand the underlying relationships within the networks and identify the root cause.

2. Problem Definition

This project aims to reduce systemic late deliveries by engineering a model that views the supply chain as a connected web rather than a list of independent events. Currently, the industry relies on spreadsheet-style math that treats every order as an independent event. This approach is fundamentally flawed because it does not consider how a single congested hub or regional disruption impacts the entire network. Furthermore, the existing smart models' black-box feature prioritizes predictive power over interpretability, neglecting the causal insights behind a failed delivery. We propose a Graph Attention Network (Veličković et al., 2017) framework - applied to Brazilian Olist E-Commerce dataset (Olist and Sionek, 2018) - to model the supply chain as a relational graph. By learning dynamic attention weights, our model captures the structural dependencies between logistics entities, providing both late delivery risk prediction and the root-cause interpretability required to mitigate bottlenecks before they escalate.

3. Literature Survey

Dwivedi et al. (2009, 2025) and Lee & Mangalaraj (2022) established the foundation of integrating descriptive and predictive analytics to enhance delivery reliability. Building on this, Lathwal et al. (2023) demonstrated ML models, specifically ensemble methods, consistently outperform traditional statistical approaches, such as ARIMA, in high-dimensional environments. Supraja et al. (2025) and Khaldy et al. (2025) reported accuracies up to 94% applying Random Forest and LSTM models to e-commerce data. While these works justify our move toward complex ML for the DataCo dataset, they suffer from a common shortcoming: they rely on simpler constraints than real-world logistics and treat each order as an isolated event (Mohamed-Iliasse et al., 2022). The independent tabular assumption does not accurately capture supply chain resilience (Spieske & Birkel, 2021) and or how risk propagates throughout the system. As Yang et al. (2023) argued, disruptions in a global network are rarely contained; they cascade from one node to another. A model that ignores these relational dependencies cannot effectively predict how a bottleneck at a single port of a distribution center might amplify risks for multitudes of other connected orders. To align flat data and network reality, Wasi et al. (2024) and Jiang (2025) proposed a paradigm shift toward Graph Neural Networks (GNNs) in the supply chain. These models create the representation of logistics as a web of interconnected nodes. However, they often rely on generalized assumptions and fail to account for unique regional constraints, like weather or infrastructure. As a result, Kipf & Welling (2016) introduced Graph Convolutional Networks (GCNs) to uniformly aggregate

adjacent elements. However, the uniform aggregation is a significant shortcoming as it cannot distinguish which specific connection, such as a congested port or failing carrier, is driving the risk. The project adopts the Graph Attention Networks (GATs) model, pioneered by Veličković et al. (2017) and explored by Wu et al. (2020), to utilize its self-attention mechanisms and dynamically weigh the influence of neighboring nodes, identifying the specific main driver node for late delivery risk. Finally, we bridge the gap between "black-box" AI and decision-making by integrating the visual frameworks of Khakpour et al. (2021) and Lou et al. (2020). Moving beyond purely descriptive dashboards, we transform internal GAT attention weights into Relational Contagion Arcs. By mapping directional links from root causes to impacted orders, we enable proactive risk mitigation, allowing us to see in real-time how bottlenecks propagate through the logistics web.

4.1 Proposed Methods: Model

By modeling the supply chain as a graph where nodes represent orders and edges represent shared logistics attributes, it replaces row-by-row pattern matching with relational intelligence. Our GAT architecture captures how delivery risk circulates through the system, separating localized delays from systemic disruptions and tracing cascading failures back to congested ports or shared carriers.

4.1.1 Restricted Spatial-Funcational k-NN

To construct the graph from tabular records without excessive computation, we apply a sparsification strategy: an edge E_{ij} exists only if two orders share the same *destination* and node j falls within the $k=5$ nearest neighbors of node i by customer coordinates. This restricts attention to peer shipments under similar geographic and contractual conditions.

4.1.2 GATv2 with Dynamic Attention

We employ GATv2 (Veličković et al.), which replaces the static neighbor rankings of standard GAT with a dynamic, non-linear attention mechanism. This allows the model to weigh the same delay differently depending on the *destination* context.

4.1.3 Residual Feature Injection

Our graph-learned insights are integrated with the original node features to incorporate relational context while preserving individual predictive signals.

4.1.4 Weighted Risk for Imbalanced Dataset

To optimize the GAT, we employ a weighted *BCEWithLogitsLoss* function, which applies a higher penalty to misclassified late deliveries, which ensures the model prioritizes the detection of high-risk late events over the more frequent on-time deliveries.

4.2 Proposed Methods: Visualization

Our visualization is a dashboard built with D3 and React, featuring three innovative and intuitive components. It uses a drill-down approach to balance aggregation and granularity, allowing the user to adjust the visualization to suit the desired level of detail.

4.2.1 Map

The map illustrates the primary innovation of the GNN model: associating orders that are related by location and time. The order location is encoded by map position, and the order time range is filtered by an interactive slider. However, simply displaying all orders is not usable because positions are determined by zip codes, each of which can contain multiple orders. This causes order markers to overlap on the map, resulting in unselectable orders. To resolve this issue, our map uses a three-level hierarchy: location, customer/seller, and order.

Assuming the *Sellers* filter is selected, the first level, *Location*, plots every location where sellers reside on the map. Here, each location marker can contain multiple sellers. When a location marker is selected, a list of sellers from that zip code is then displayed under the next level: the *Sellers* panel. Additionally, a graph is overlaid on the map, illustrating all the locations that the chosen seller delivered to within the time window. This feature gives stakeholders an intuitive picture showing both shipping distance and seller workload (the number of orders the seller fulfilled in the window). When a specific seller is selected from the *Sellers* panel, all of their orders are displayed in the final level: the *Orders* panel. Lastly, when an order is chosen from the *Orders* panel, the overlaid graph shows an edge connecting the seller's location and the order's destination, as well as showing the *Delivery Timeline* and *Related Orders* panels.

The map has multiple filtering options for customization. Users can toggle between the customer or seller views, but both cannot be viewed simultaneously due to overlapping issues from customers and sellers. There is also an option to filter by delivery outcome: either on time, late, or both. Finally, the map can be filtered by a weekly time range. We use a slider with a fixed window instead of a standard calendar with selectable dates for three reasons. First, displaying all locations on the map at once is extremely slow, so enforcing a maximum interval is crucial. With the constraint, the calendar-based UI becomes confusing because the second date cannot be chosen freely, making a time slider a better choice. Second, temporal relationships between orders like weather and congestion are typically relevant for only within a three to seven day window. Therefore, larger windows introduce noise to our chart, while smaller windows are too sparse to yield meaningful insights. Finally, *weeks* are a more intuitive unit for stakeholders compared to arbitrary 3-day or 5-day intervals.

Lastly, based on usability testing, we added legends to every chart. Sellers are encoded with squares (both start with "S"), and customers are represented by circles (both start with "C"). Additionally, color is used to indicate delivery outcomes.

4.2.2 Delivery Timeline

Our delivery timeline is the most straightforward chart, but it is crucial for a delivery-time predictor. It is a Gantt Chart that splits orders into phases along a timeline, allowing stakeholders to easily identify bottlenecks. It uses length to encode time and color to indicate phases that are behind schedule.

4.2.3 Visual Model Representation

The visual model representation, or *Related Orders* chart, describes how our black-box model operates under the hood. It uses an intuitive graph structure to display an order and its k-nearest neighbors,

mirroring the exact algorithm our model uses to make predictions; it displays the selected order’s five nearest neighbors because we found that the model’s performance peaked when k is 5.

Here, we apply KNN based only on customer location, not seller location, because sellers are highly likely to have multiple orders in a given time window. As a result, an order’s nearest neighbors would often share the same seller, rendering the graph feature of our model useless since this information is easily captured with the original tabular dataset. Additionally, this would introduce noise into the dashboard. This issue does not apply to customers, who are less likely to place many orders in a short period, so an order’s nearest neighbors typically come from nearby customers.

A tooltip is added to each node to display additional order information, including the distance between orders and the date they were placed. These two features are highlighted to emphasize the innovations of our graph-based prediction model.

5. Experiment Design & Evaluation

We evaluate the dataset through two distinct pipelines:

- **Tabular Baseline Pipeline:** Established to benchmark pure predictive performance. This track utilizes **Random Forest**, **XGBoost**, and **MLP** models. The interpretability is derived through standard feature-importance analysis.
- **Graph Learning Pipeline:** Implemented using PyTorch Geometric, this track tests whether capturing the underlying logistics network, defined by shared geographic information, provides a predictive advantage by comparing **GCN**, **GNN**, and **GAT** architectures.

5.1 Dataset

We originally performed experiments with the DataCo Smart Supply Chain dataset (Constante et al., 2019). However, upon further analysis, we discovered that the dataset has a high likelihood of being synthetic, which was not clarified in the dataset’s description; when we visualized the distribution of order delivery times, we discovered a uniform distribution, which does not align with the normal or right-skewed distribution that is commonly seen in delivery-time data. Furthermore, all 20,000+ instances of first-class orders were delivered in exactly two days, which is highly suspicious.

In order to develop a model and visualization that draws meaningful insights, we switched to the Brazilian E-Commerce dataset (Olist & Sionek, 2018), which is confirmed to be real supply-chain data. It provides a strong empirical foundation because it supports end-to-end analysis of actual e-commerce operations, including customer behavior, order activity, product categories, and delivery performance.

5.2 Feature Selection

We selected features that reflect four main aspects of delivery risk: seller identity, product attributes, payment method, and geographic delivery context. The final feature set included seller ID, product category, product weight, product length, product height, product width, seller city, customer city, payment type, seller and customer latitude and longitude, and a derived seller-to-customer distance variable. These features were chosen to capture the characteristics of the product being shipped, the locations involved in fulfillment, and the spatial conditions that may affect delivery timeliness.

In preprocessing, the payment type was one-hot encoded, while seller ID, seller city, customer city, and product category were target encoded. The continuous variables, including the product-dimension features and the distance variable, were scaled, and the geographic coordinates were normalized to support stable training across both tabular and graph-based models.

5.3 Model Architecture

Our core architecture is a Relational-Tabular Hybrid, designed to simultaneously capture individual order characteristics and systemic network dependencies. We employ a GATv2Conv layer with eight attention heads for dynamic neighbor weighting, paired with Batch Normalization to stabilize training, and Residual Skip Connections to mitigate oversmoothing. A final predictive head concatenates these graph-learned embeddings with raw tabular features, ensuring the model preserves individual signals like destinations while leveraging broader network context.

To validate our attention mechanism, we benchmarked the GAT against a Graph Convolutional Network (GCN) using a uniform neighborhood weight and a basic Graph Neural Network (GNN) that performs simple neighborhood-based message passing without the explicit attention mechanism that distinguishes GAT. These controls isolate performance gains attributable to dynamic attention versus structural connectivity alone. By training all models on identical node features, graph constructions, and train-test splits, this standardizes the comparison of each model’s ability to prioritize logistics bottlenecks.

5.4 Model Evaluation

To validate the efficacy of our graph-based approach, we benchmarked our GAT architecture against a suite of standard tabular ensemble methods and alternate graph configurations. The experiments were conducted using an 80/20 train-test split, with performance measured across Accuracy, weighted F1 Score, and AUC-ROC.

Model	Accuracy	(Weighted) F1-Score	AUC-ROC
Random Forest	94%	92%	74%
XGBoost	78%	83%	71%
GCN	86%	88%	69%
GAT	74%	80%	69%
GNN	87%	88%	70%
MLP	86%	88%	69%

The **Random Forest** baseline achieves the highest metrics (92% F1, 74% AUC-ROC), confirming that tabular features like seller ID and customer city are highly predictive when treated as independent components. Our **GNN** and **GCN** models demonstrate strong performance (88% F1), effectively outperforming the XGBoost baseline. This validates our hypothesis: by constructing graph edges based on shared seller ID and customer city attributes, we successfully introduce structural context that traditional models lack. The **GAT** (80% F1) trails the top ensemble methods in raw classification performance. However, this is an intentional diagnostic trade-off. Unlike the black-box Random Forest, the GAT’s

attention mechanism allows us to map the influence of specific nodes, effectively highlighting why a seller-city pairing is at risk.

5.5 Visualization Evaluation

To evaluate our visualization, we conducted usability testing in two phases: **task execution** and **insight confirmation**. In the task execution phase, participants completed targeted tasks across each panel. Next, in the insight confirmation phase, we checked whether the perception of the data was accurate.

For the map, participants identified the seller with the most late orders in a given location and the number of locations containing late orders within a specified period. For the delivery timeline, participants determined which phase exceeded its expected duration the most and which phase had the shortest actual duration. For the Related Orders panel, they identified the single nearest neighbor to a given late order and the percentage of nearest neighbors that shared the same delivery outcome.

Key takeaways from our testing included incorporating legends and separating the customer view from the seller view. We initially assumed the visualization would be intuitive without legends, but participants suggested adding keys for greater clarity. Additionally, we found that seller squares and customer circles became incomprehensible when they overlapped, so we separated the two views.

6. Conclusions & Discussion

Although tabular models achieved higher predictive performance by treating orders as isolated events, graph-based architectures are beneficial for capturing systemic supply chain contagion and providing the structural foundation necessary to visualize how disruptions propagate throughout the network. This highlights a strategic trade-off: tabular models optimize for classification, while GAT-based frameworks prioritize root cause diagnostics and structural interpretability.

The performance gap also suggests that geographic k-NN proximity alone is an insufficient proxy for complex operational dependencies. Future refinements should expand edge definitions to include temporal flows and shared distribution hubs, moving beyond static spatial data. Additionally, model robustness could be enhanced by shifting to granular multiclass classification that distinguishes between minor delays and systemic failures, and integrating external factors like seasonality and weather, which are likely more indicative of network latency than static geographic coordinates alone.

— All team members have contributed a similar amount of effort —

References

- [1] Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Yu, P. S. (2020). A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems*, 32(1), 4-24.
- [2] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2017). Graph attention networks. *arXiv preprint arXiv:1710.10903*.
- [3] Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*.
- [4] Jiang, Y. (2025, March). Artificial intelligence driven supply chain network optimization: An empirical analysis based on reinforcement learning. In *Proceedings of the 2025 4th International Conference on Cyber Security, Artificial Intelligence and the Digital Economy* (pp. 244-247).
- [5] Wasi, A. T., Islam, M. D., Akib, A. R., & Bappy, M. M. (2024). Graph neural networks in supply chain analytics and optimization: Concepts, perspectives, dataset and benchmarks. *arXiv preprint arXiv:2411.08550*.
- [6] Mohamed-Iliasse, M., Loubna, B., & Abdelaziz, B. (2022). Machine learning in supply chain management: A systematic literature review. *International Journal of Supply and Operations Management*, 9(4), 398-416.
- [7] DWIVEDI, V., Kadian, K., Gardiner, B., Pandey, R., & Lathwal, A. (2025). Supply Chain Analytics: A Performance Evaluation of Machine Learning, Statistical, and Time-Series Models. *Baltic Journal of Modern Computing*, 13(2).
- [8] Lee, I., & Mangalaraj, G. (2022). Big data analytics in supply chain management: A systematic literature review and research directions. *Big data and cognitive computing*, 6(1), 17.
- [9] Lou, C. X., Bonti, A., Prokofieva, M., Abdelrazek, M., & Kari, S. M. C. (2020, September). Literature Review on Visualization in Supply Chain & Decision Making. In *2020 24th International Conference Information Visualisation (IV)* (pp. 746-750). IEEE.
- [10] Khakpour, A., Colomo-Palacios, R., & Martini, A. (2021). Visual analytics for decision support: A supply chain perspective. *IEEE Access*, 9, 81326-81344.
- [11] Al Khaldy, M. A., Alnaimat, F., Al-Qerem, A., Ali, A. M., & Rajab, M. (2025, April). Towards smarter supply chains: Machine learning methodologies for delivery risk assessment. In *2025 1st International Conference on Computational Intelligence Approaches and Applications (ICCIAA)* (pp. 1-7). IEEE.
- [12] Supraja, N. B. N., Pujahari, O. P., Singh, N., & Sahoo, S. K. (2025). Optimizing Supply Chain Management through Machine Learning: Applications in E-Commerce. *Journal of Marketing & Social Research*, 2, 235-244.
- [13] Yang, M., Lim, M. K., Qu, Y., Ni, D., & Xiao, Z. (2023). Supply chain risk management with machine learning technology: A literature review and future research directions. *Computers & Industrial Engineering*, 175, 108859.
- [14] Spieske, A., & Birkel, H. (2021). Improving supply chain resilience through industry 4.0: A systematic literature review under the impressions of the COVID-19 pandemic. *Computers & industrial engineering*, 158, 107452.
- [15] Dwivedi, A., et al. (2009). Analytics in supply chain management: Enhancing performance through structured data integration. *Baltic Journal of Modern Computing*, 13(2). https://joscm.refconf.com/article_2813_e7e7abd9d8f4046cd8c0970ffe1c7332.pdf

- [16] Constante, F., Silva, F., & Pereira, A. (2019). *DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS* (Version 5) [Data set]. Mendeley Data. <https://doi.org/10.17632/8gx2fvg2k6.5>
- [17] Olist, and André Sionek. (2018). Brazilian E-Commerce Public Dataset by Olist [Data set]. Kaggle. <https://doi.org/10.34740/KAGGLE/DSV/195341>
- [18] Xu, Zhenyu, Yimin Tan, Xinxue Wang, and Xiang Li. “Data-Driven Analysis for the Operation Status of the e-Commerce Platform Based on Olist.” <https://www.scitepress.org/Papers/2022/118366/118366.pdf> *Proceedings of the 1st International Conference on Public Management, Digital Economy and Internet Technology (ICPDI 2022)*, SCITEPRESS, 2023, pp. 859–867.
- [19] Ye, Jingyi. “Analysis on E-commerce Order Cancellations Using Market Segmentation Approach.” <https://dl.acm.org/doi/fullHtml/10.1145/3450588.3450596> *Proceedings of the 2021 4th International Conference on Computers in Management and Business (ICCMB 2021)*, Association for Computing Machinery, 2021, pp. 33–40.